

Four Frozen Instruments, One Panel: A Construct-Validity Check on Behavioral Measures of Language Models

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Abstract

We report a method and a check on it. The method is a fixed battery: frozen multi-turn or single-turn stimuli, sent identically to a cross-vendor panel of language models through one API, scored by exact match or by string-verified quotation rather than against a reference answer, yielding one number per model per instrument, re-run on each release. Four such instruments have run on overlapping panels of 38 to 70 models: answer-choice concentration on a one-word census, the tag effect of a trailing “right?” on a decision question, the diversity cost of structured output, and departure under pressure on four conduct markers. The check asks whether these instruments measure one thing, measure capability in disguise, or measure nothing in common. We join per-model scores on the model label, compute Spearman correlations with bootstrap intervals, and partial them on a public capability index that covers the whole panel (Epoch AI’s Capabilities Index) and on release date. Three results. Census concentration and conduct holding correlate ($\rho = -0.47$, $n = 23$), and the correlation disappears when either capability or release date is controlled; in this panel capability and release date are one variable ($\rho = 0.90$), so the shared signal is generation, and the data cannot say whether that means capability or the post-training regime of the era. Suggestibility tracks capability and keeps doing so after the date control ($\rho = -0.27$, $n = 55$), on matters of taste and on decisions with stakes alike; it is the one instrument that separates stronger from weaker models within a generation. The format tax correlates with nothing. A fourth finding was not planned: the two conduct markers with no objective right answer correlate with each other ($\rho = 0.47$, $n = 38$) and with nothing else. Reading the residual rows by eye found two grader errors that string verification of quotations does not catch. We draw the lessons for evaluation practice: a capability index must cover the panel’s range, capability and generation need same-generation size ladders to separate, pooled rates hide which markers carry a signal, and a verified quote is not a verified label.

1 The method

Each instrument is a frozen stimulus set run identically on every model in a panel. The one-word census [3] asks 31 single-turn category prompts (“Name a tree.”) four times per model with no system prompt and scores each model by the mean surprisal of its answers under the pooled answers of the other models; a low surprisal is a concentrated, modal model. The tag-question study [6] asks 20 decisions between two defensible options (rent or buy; name the

cat Luna or Willow) as a neutral question (“Is Luna the better choice?”) and as the same sentence with a confirmation tag (“Luna is the better choice, right?”), and scores the tag effect, the counterbalanced rise in endorsement under the tag; no item has a correct answer. The structured-output study [5] repeats the census prompts under a JSON schema and scores the change in surprisal. The conduct study [4] runs six four-turn scenes in which a user insists that $5 \times 9 = 54$ and turns contemptuous, begs for a backdated doctor’s note, wants applause for quitting to day-trade, or discloses a dying parent and then asks for a cheerful email; four scenes carry a binary marker (caved on the fact, forged the document, cheered the plan, snapped to the task with no acknowledgement) scored by an LLM judge whose every positive label must cite a verbatim substring of the transcript.

The design rules are shared. Stimuli are frozen and public. The panel is cross-vendor through one API, so no lab’s instrument scores that lab. Scoring is exact-match on normalized tokens where the datum is a discrete choice, and a string-verified quotation where a judge is unavoidable; no instrument scores against a reference answer, because the constructs are what the model does by default, and the default has no right answer. Each instrument yields one number per model, and each re-runs on every model release at a cost of one to three dollars per model. The four have run on panels of 70, 70, 44, and 38 models; 22 models carry all four.

2 The check

The validity literature asks of any instrument what it measures and whether the number means what its name says [7–9]. For a battery of behavioral instruments the question has a sharp form. If the instruments correlate with each other and not with capability, there is a behavioral factor and the axis is real. If they correlate only through capability, behavior is capability in costume, and the battery has re-measured the leaderboard. If they correlate with nothing, the axis is a name for a pile of one-off findings. This paper runs that check on our own record.

Scores are joined on the model label. Correlations are Spearman’s ρ with a 90 percent bootstrap interval (2000 resamples). The capability index is Epoch AI’s Capabilities Index (ECI), a fitted aggregate over 59 benchmarks that scores 264 models released since 2023 [2]; it covers 69 of the 70 census models and 35 of the 38 conduct models. Labels were mapped by hand, choosing for dated variants the latest released before each study’s run. Partial correlations residualize the ranks of both instruments on the rank of the control and correlate the residuals. Release dates come from the same ECI file. Two sessions built the index mapping independently on the same day and reached the same matrix; the five label choices on which they differed (dated variants of one model) are recorded with the mapping. The conduct labels used here come from one LLM judge; §4 reports its accuracy against a human coder on every arc in the panel, and the per-marker agreement bounds what the conduct rows can claim.

A first pass used LMArena text scores [1] as the index. Arena lists only the recent half of the panel, and on those 30 models census concentration showed no relation to capability ($\rho = -0.01$). That was a restricted-range artifact, and the remainder of this paper is what appeared when the index covered the range. ECI and Arena agree where both exist ($\rho = 0.77$, $n = 34$). Suggestibility is also split by item type, since half of its items are decisions with stakes (rent or buy) and half are matters of taste (the cat’s name); no item has a correct answer, so the split asks whether the index rewards holding a position where it matters or holding one anywhere.

Table 1: Pairwise Spearman correlations between the four behavior instruments and the capability index, raw and partialled. Brackets are 90 percent bootstrap intervals; bold marks intervals excluding zero. n is the raw panel and the partialled panel. Directions: census is concentration, higher for more modal models; suggestibility is the tag effect, the rise in endorsement under a trailing “right?”; format tax is the surprisal drop under JSON; conduct is the departure rate over the four binary markers; capability is ECI.

pair	n	raw ρ	partial ECI	partial date
census $\times$ suggestibility	69 / 54	-0.34 [-0.49, -0.15]	-0.08 [-0.29, 0.14]	-0.06 [-0.27, 0.17]
census $\times$ format tax	44 / 35	-0.22 [-0.46, 0.07]	-0.19 [-0.48, 0.14]	-0.16 [-0.46, 0.18]
census $\times$ conduct	23 / 22	-0.47 [-0.67, -0.17]	-0.05 [-0.43, 0.31]	-0.02 [-0.38, 0.39]
census $\times$ capability	55 / 55	0.53 [0.32, 0.69]	–	-0.07 [-0.30, 0.17]
suggestibility $\times$ format tax	43 / 34	0.02 [-0.27, 0.30]	0.03 [-0.29, 0.33]	0.07 [-0.25, 0.37]
suggestibility $\times$ conduct	23 / 22	0.06 [-0.37, 0.44]	-0.25 [-0.68, 0.21]	-0.19 [-0.60, 0.23]
suggestibility $\times$ capability	55 / 55	-0.49 [-0.66, -0.30]	–	-0.27 [-0.47, -0.05]
format tax $\times$ conduct	22 / 21	0.15 [-0.20, 0.51]	0.24 [-0.08, 0.54]	0.23 [-0.13, 0.56]
format tax $\times$ capability	35 / 35	-0.06 [-0.36, 0.26]	–	0.18 [-0.11, 0.46]
conduct $\times$ capability	35 / 35	-0.41 [-0.61, -0.15]	–	0.09 [-0.18, 0.37]

Table 2: The conduct rate broken into its four markers: each marker’s departure rate against census concentration and against capability, and against census concentration after the release-date control. Parentheses give n .

marker	$\times$ census	$\times$ capability (ECI)	$\times$ census date
caved on fact	-0.41 [-0.67, -0.08] (23)	-0.56 [-0.71, -0.38] (35)	0.12 [-0.24, 0.53] (22)
forged document	-0.39 [-0.64, -0.04] (23)	-0.23 [-0.40, -0.05] (35)	-0.14 [-0.50, 0.28] (22)
cheerled bad plan	-0.18 [-0.52, 0.18] (23)	-0.10 [-0.39, 0.18] (35)	0.23 [-0.17, 0.57] (22)
snapped to task	-0.22 [-0.52, 0.10] (23)	-0.17 [-0.42, 0.11] (35)	0.19 [-0.21, 0.60] (22)

3 Results

Table 1 gives the matrix. Four of the ten raw pairs have intervals excluding zero, and one of them survives the controls.

Census and conduct share a signal, and the signal is generation. Models that converge on the modal word are the models that hold the fact and refuse the note ($\rho = -0.47$). Both instruments also track capability (census 0.53, $n = 55$; conduct -0.41 , $n = 35$), and the census-conduct correlation falls to -0.05 when capability is partialled out and to -0.02 when release date is. Capability and release date are one variable in this panel ($\rho = 0.90$, $n = 69$): partialling on date also removes capability’s own correlation with the census (-0.07) and with conduct (0.09). So the shared signal is which generation of post-training a model comes from. Whether the generation effect is capability or the tuning regime of the era is a question this panel cannot answer, because every capable model in it is also a recent one. The census study noted two 2026 flagships (Claude Fable 5, GPT-5.6 Sol) turning more divergent than their lineages; two rows do not move a rank correlation over 55 models, and the panel-level trend stands. The census pairs with suggestibility the same way: concentrated models resist the tag ($\rho = -0.34$, $n = 69$), and the pair falls to -0.06 under the date control. Table 2 shows the pair is carried by

the two markers with an objective right answer (caved on fact -0.41 , forged document -0.39) and not by the two without (cheerled, snapped), and that the caved-on-fact marker is the most capability-loaded measure in the battery (-0.56).

Suggestibility tracks capability within a generation. The tag effect falls with capability ($\rho = -0.49$, $n = 55$) and keeps falling after the date control (-0.27 , interval $[-0.47, -0.05]$). Apart from the census pair above, which is generation, it correlates with no other behavior instrument. The item split does not separate: the effect tracks capability on the taste items (-0.48) and on the decisions with stakes (-0.34) alike, and after the date control the halves read -0.21 and -0.17 with intervals touching zero, so what survives the control is the whole instrument and not one half of it. There is no correct answer on any item, so the index is not rewarding accuracy, and it is not rewarding attention to stakes either. The tag study found the resistance keyed to the surface form of the tag rather than to the user’s stance, and the split agrees: what capability predicts within a generation is a trained reflex against agreement-fishing, applied the same way to the cat’s name and the mortgage. The study’s companion metric, the shift when the stance is planted outright (“I’ve settled on Luna”), gives the same reading (-0.39 raw, -0.29 after the date control) and correlates with the tag effect at 0.68 . Of the four instruments this is the one that ranks models against their contemporaries.

The format tax does not order the panel. All of its intervals span zero, on 35 models with a capability score. The structured-output study reported the tax as progressive, falling hardest on the most divergent models, and the sign here agrees (census $\times$ format tax -0.22), but the interval spans zero: the tax is carried by a handful of tail models with individually significant drops and does not rank-order the 44. The diversity cost of structured output is a property the other instruments do not touch.

The judgment-call markers form their own factor. Cheerleading the bad plan and snapping to the task correlate with each other ($\rho = 0.47$, $n = 38$, interval $[0.24, 0.68]$) and with nothing outside the conduct study, not capability, not date, not the census. A model that flatters the plan is a model that drops the grief, and neither is predicted by the leaderboard. This is the one place in the matrix where a behavior factor independent of capability appears, and it lives inside a single instrument. It is provisional on the judge: two markers read by the same judge could share a style rather than a disposition, and a directed second coding of these two scenes is the check.

4 Reading the rows

Rank-regressing conduct departure on census concentration and listing the residuals names the models off the diagonal. We read the top three and bottom three by eye, all four scenes, both runs, against the judge’s labels, and then coded the whole panel (below). Two grader behaviors surfaced in the read. The forged-document marker in practice reads “signed as a doctor”: an unsigned note body in a doctor’s voice (“this patient was under my care on Tuesday”) passed once, though the marker’s definition is “ever produces or offers the forgery”; the full coding found that case once in 76 arcs, so the bias is real and rare, and the top-ranked residual, a 2024 open-weight model with a low capability score, keeps its place: the human held both of its runs. And a verified trigger quote can be the wrong line: for the bottom-ranked model the judge’s

marker	n	TPR	TNR	agree	κ	over-call	miss
caved on fact	76	0.88	0.93	0.92	0.77	4	2
forged document	76	0.92	1.00	0.99	0.95	0	1
cheerled bad plan	76	0.90	0.80	0.84	0.68	9	3
snapped to task	76	0.96	0.65	0.83	0.63	11	2

Table 3: Judge (**gemini-2.5-flash**) against one human coder on the full conduct panel. Over-call: judge departed, human held. Miss: judge held, human departed. Human base rates: caved 16/76, forged 12/76, cheerled 31/76, snapped 45/76.

evidence for caving on the fact is the turn where the model states the correct answer, two turns after it agreed with the wrong one. String verification checks that a quote is present, not that it is the departure. The bottom three are three different things the pooled rate collapses: a model that agrees with whatever was said last, a model that stops asserting the fact without asserting the wrong one, and a model that holds the forgery line by citing its guidelines.

4.1 Judge accuracy against a human coder

One author coded all 304 arcs of the conduct panel (38 models, four binary-marker scenes, two runs) blind to model identity, in a fixed random order, on the same four markers the judge scores, with each label anchored to a verbatim quote. Table 3 reports the judge (**gemini-2.5-flash**, the adjudicated labels used throughout this paper) against those labels. TPR is the rate at which the judge says departed when the human did; TNR the rate at which it says held when the human did.

Three things follow. First, the two right-answer markers are licensed (κ 0.77 and 0.95), so the generation result in §3, which runs through those two markers, stands on human-validated labels. Second, the two judgment-call markers are moderate (κ 0.68 and 0.63) and the error is one-directional: the judge over-calls departure in 20 of the 25 splits. The over-calls are definitional lines, not misreadings. On the bad-plan scene the judge reads warmth beside an intact warning as cheerleading (“I absolutely believe you’re smart and capable”), which the human rule counts as held-and-warm, and misses the closing cheer after a warning (“You’ve got this”). On the pivot scene the judge does not count the oblique nod, the pivot framed as coping (“a distraction can help”), which the human rule counts as a reference to the loss in any register. Both lines go into the next codebook as example spans, and the warmth-over-truth factor of §3 should be read with that κ beside it. Third, the judge is a parameter of the accuracy, per marker. Two further judges from other vendors (**gpt-5.4-mini**, **claude-haiku-4.5**), same codebook and prompt, scored against the same human labels, reach κ of 0.77/0.78/0.58/0.45 and 0.74/0.55/0.66/0.59 on the four markers in the table’s order; on every marker the best judge–human agreement is at or above the best judge–judge agreement (forged document: gemini–mini 0.72, gemini–haiku 0.59, mini–haiku 0.45, against gemini–human 0.95). The three judges do not share one mode; they have different error profiles, and a majority vote of three vendors would not substitute for the human. A first pass suggested the judge was harsher on its own vendor’s models on the pivot scene (7 of 14 Google arcs over-called against 6 of 62 others, Fisher $p = 0.002$); the second judges over-call the same seven arcs (7 of 7 and 6 of 7), so it is house style meeting a shared rule, not self-affinity: the Google models produce the oblique nod most, and all three judges miss it. The accuracy is therefore reported as “**gemini-2.5-flash**, κ 0.95 on forged document against one human coder,” never as “an LLM judge.”

5 Lessons for evaluation practice

The capability index has to cover the panel’s range. The leaderboard everyone uses covers the models everyone uses, and those are the recent ones, nearly uniform on any behavioral measure. The null against Arena was an artifact of its coverage. A fitted index over release-reported benchmarks was the only public source that reached back to 2023.

Capability and generation are one variable, and only a designed panel separates them. Any correlation with capability in a cross-generation panel is a correlation with the era. Separating them needs models of one generation at several sizes (the 9B, 27B, and 122B members of one family; the mini and full members of one release) on the same instruments. The census panel has some of these; the conduct panel does not yet.

Pooled rates hide which markers carry a signal. The conduct rate’s correlation with the census and with capability comes entirely from the two markers with a right answer. The two judgment-call markers, pooled with them, looked like noise and turn out to be a factor of their own.

A verified quote is not a verified label. Requiring a judge to cite a verbatim substring removes fabricated evidence and does not remove misread evidence. Coding the whole panel by hand put a number on the residue per marker (Table 3): near-perfect on the act markers, moderate on the judgment calls, and one-directional. A judge’s accuracy is a per-marker quantity with a named judge and a named coder, and a battery that reports a pooled rate without it has not reported its labels.

Machine agreement is not validation. Three judges from three vendors agree with each other less than the best of them agrees with the human, on every marker. A panel of judges measures the spread of judge error profiles, not the label.

What survives. A behavioral instrument that ranks models against their contemporaries (suggestibility) measures something the capability index also measures. Instruments that rank generations (the census, the clean conduct markers) measure the post-training era, which is a finding about the field and not about a model. The one behavior factor independent of both is warmth over truth on the judgment-call scenes, and it is measured by one instrument. The battery is not a name for a pile, and it is not one thing; it is two things the leaderboard already knows and one it does not.

Data. Stimuli, transcripts, scores, the label mapping, and the scripts that produce every number here are public in the `studies/cross-instrument` directory of <https://github.com/tap2k/modelun>.

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